

Neural Networks & Deep Learning (CSC-430)
Midterm Exam
Date: 20-November-2025
Name:

Instructor: Dr. Shamweel Qamar
Max Marks: 50
Time Allowed: 60 mins
Roll #:

- Attempt the paper with blue or black pen, question attempted with pencil will be considered as rough work and will not be marked
- Attempt ALL questions on the answer booklet, questions attempted on the Question paper will not be marked

[CLO-1] Q1: Describe the role of each component in a fully connected neural network (weights, biases, activation functions). (Marks: 10)

[CLO-1] Q2: Differentiate between batch gradient descent and stochastic gradient descent (SGD) citing their pros and cons in terms of computational efficiency and convergence. (Marks: 5)

[CLO-3] Q3: You are working in a company as a Machine Learning Engineer. The company is asking you to create an email classification system for them. Explain in detail what are the steps you are going to take to solve this problem. (Marks: 15)

[CLO-2] Q4: Now, imagine that you are going to work on an image classification task. Which neural network architecture you will prefer and why? Explain your reasons. (Marks: 10)

[CLO-1] Q5: Draw the architectural diagram of the neural network you are going to use for the binary image classification task (cat vs human). Describe the name of each layer and its purpose. (Marks: 5)

[CLO-1] Q6: As a Machine Learning Engineer, you got to know that the model you have trained is overfitting. What options are you going to explore? Enumerate and explain. (Marks: 5)

CLO	Description of CLOs	Mapping to Questions
CLO1	Understand and explain the theoretical foundations of deep learning	Q1, Q2, Q5, Q6
CLO2	Apply neural network models using deep learning frameworks	Q4
CLO3	Develop/Construct deep learning techniques to solve real-world problems	Q3

-----End of exam-----